Exercise 3.3

1. cs= 400 units
2. C.S=378 units
3. C.S=562.50 units
4. C.S=$\frac{1}{2}\left[1-\log_e 2\right]$ units
5. P.S=$\frac{25}{2}$units.
6. P.S=237.3 units
7. $C.S = 36\log\frac{3}{2} - 12\,units$
8. $\frac{32000}{3}$units
9. C.S=$\left(8\log 2-4\right)$units, P.S=$\frac{1}{4}$units
10. C.S=$\frac{1024}{3}$units, P.S=64 units
11. C.S=24 units, P.S=16 units

Exercise 3.4

1	2	3	4	5	6	7	8	9	10	11	12	13
(c)	(b)	(a)	(c)	(a)	(a)	(d)	(c)	(b)	(a)	(a)	(a)	(b)
14	15	16	17	18	19	20	21	22	23	24	25	
(c)	(c)	(b)	(a)	(b)	(a)	(c)	(a)	(c)	(a)	(b)	(c)	

Miscellaneous problems

1. ₹1,900
2. C= ₹3,125
4. $R = 6x - x^3 - \frac{x^4}{4}$, $p = 6 - x^2 - \frac{x^3}{4}$
5. Profit function is $10x - \frac{x^2}{40} - 100$.
6. $C.S = \frac{40}{9}$ units, $P.S = \frac{32}{9}$units.
7. 52,770 units
8. $P = 11 - \frac{x^3}{3}$
9. $\frac{76}{3}$ sq.units
10. $\frac{1}{5}\left[(2)^{\frac{5}{3}} - 1\right]$ sq.units

4. Differential Equations

Exercise 4.1

1.(i) (1 , 1) (ii) (3 , 1) (iii) (2 , 2) (iv) (3 , 1)
(v) (3 , 3) (vi) (2 , 1) (vii) (1 , 4).

2.(i) $y = x\frac{dy}{dx} + \frac{dy}{dx} - \left(\frac{dy}{dx}\right)^2$ (ii) $\left(\frac{dy}{dx}\right)^3 - 4xy\frac{dy}{dx} + 8y^2 = 0$ (iii) $y + x\frac{dy}{dx} = 0$

(iv) $x + y\frac{dy}{dx} = 0$
3. $r^2\left(\frac{d^2y}{dx^2}\right)^2 = \left[1+\left(\frac{dy}{dx}\right)^2\right]^3$
4. $y = x\frac{dy}{dx}$
5. $2a\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^3 = 0$
6. $y^2 = x^2 + 2xy\frac{dy}{dx}$
7. $y = 2x\frac{dy}{dx} + y\left(\frac{dy}{dx}\right)^2$

Exercise 4.2

1.(i) $e^{-y} + ax + C = 0$ (ii) $\log x + \frac{x^2}{2} = \frac{y^2}{2} + \frac{y^3}{3} + C$
2. $\log x - x = \log y + C$
3.(i) $x = Cy$ (ii) $\log(1+y) = -e^x + C$
4. $(1+\sin x) = C(1+\cos y)$

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